T800 Simulation & Control Issue Report

> **To:** EngineAI Technical Support
> **From:** Guilherme Andrade (Zush)
> **Subject:** MuJoCo Simulation Issues: Missing Commands and Stand-up/Stability Failures

Hello EngineAI team,

I am reaching out to report a behavioral issue with the T800 robot within the MuJoCo simulation environment. While developing a Reinforcement Learning policy, I noticed that the simulated robot struggles significantly to stand up properly and maintain basic stability.

Furthermore, it appears that several control commands/actuation parameters that should be available to correct this behavior are missing or not properly mapped in the provided MuJoCo environment compared to the physical hardware expectations.

1. The Core Issue (MuJoCo Simulation)

- **Stand-up Failure:** The T800 model in MuJoCo exhibits structural difficulties in getting up and sustaining a stable upright posture, often triggering early termination due to a low center of mass.
- **Missing Commands:** The control interface provided for the simulation lacks specific joint commands or balancing parameters that would normally allow the controller to stabilize the torso and legs effectively.
- **Attachments Provided for Review:**
    - **Simulation Videos:** Showing the exact moments the robot fails to stand up.
    - **Training Logs:** TensorBoard statistics (PPO step data, reward, and value loss).
    - **Datasheet & Statistics:** Extracted metrics from my runs.

2. Host System Specifications

- **Operating System:** Linux Mint
- **File System:** ext4
- **Primary User Path:** `/home/guilerme-andrade/`
- **Network IP Range:** `192.168.0.0/24` (Robot is at `192.168.0.162`)

3. EngineAI SDKs & Workspaces

- **EngineAI Robotics Native SDK:** `~/engineai_robotics_native_sdk/` (Asset: `assets/resource/t800.xml`)
- **EngineAI ROS2 Workspace:** `~/source/engineai_workspace/`

4. Reinforcement Learning Stack (Python)

- **Gymnasium:** `gymnasium` and `gymnasium.envs.mujoco`
- **Stable-Baselines3:** PPO algorithm (`MlpPolicy`).

5. Docker Environment Configuration

I am running the simulation and training inside a Docker container using a highly parameterized setup. I have compressed and attached my entire Docker configuration context for your review, as the base image and variables are injected dynamically at runtime.

- **Network Mode:** `host`
- **Attached File:** `t800_docker_setup.tar.gz`

Could your engineering team review the attached videos and logs alongside the `t800.xml` configuration? I would appreciate any guidance on whether there is an updated MuJoCo model, a patch for the missing commands, or specific PD gains/actuator limits I need to adjust for the T800 to stand up correctly.

Thank you,
Guilherme Andrade